

Probability is all around us...

Everyday Life: weather, traffic, diet and exercise... Games: rock-paper-scissors, the lottery, dice, roulette, cards... World Events: elections, polls, life expectancy, natural disasters... Inherited Traits: hair type/color, eye color, blood type, dimples, freckles, diseases...	What is probability? Frequentist definition: → Outcomes are governed by chance, but in many repetitions <u>so a pattern emerges</u> → Mathematics helps understand these regular patterns A random (stochastic) process is one where we're aware of all the possible outcomes but have no idea what the next outcome will be.
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Sample Space: ALL of the possible outcomes

Event: A specific, desired outcome or set of outcomes that is a *subset of the sample space.

Notation: Sample Space $\rightarrow \Omega = \{...\}$ Probability of Event A $\rightarrow P(A)$

Range of Probabilities: The probability of an event is between 0 and 1. $[0,1]$

Sum of Probabilities: The probability of the whole sample space is 1.

We discussed the differences between theoretical probability and empirical probability in our intro activity:

Theoretical Probability

- What should happen given the sample space



Expectation

vs



Reality

$$\frac{\text{Number of Desired Outcomes}}{\text{Number of Possible Outcomes}}$$

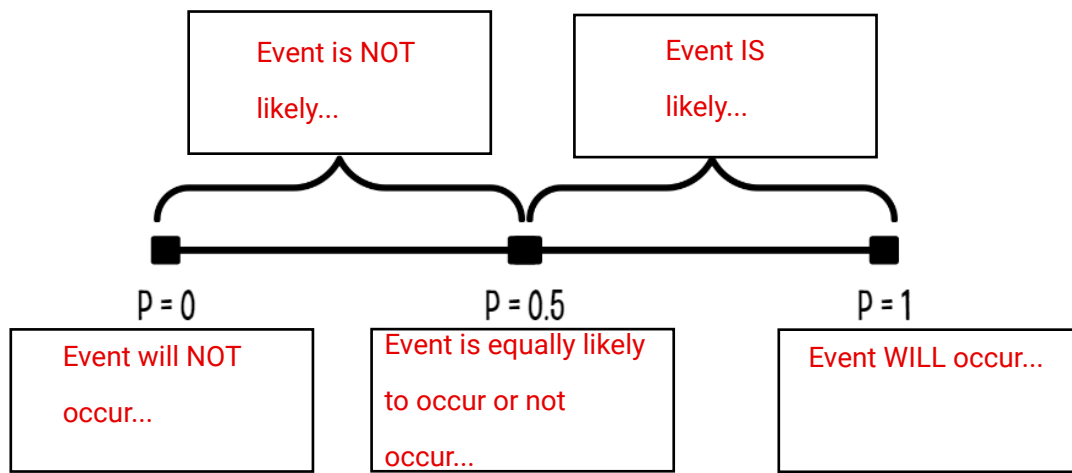
Empirical Probability

- When you are performing a simulation or experiment, it is what does happen given the trials

$$\text{Number of Successes} / \text{Number of Trials}$$

The Law of Large Numbers states that simulated (empirical) probabilities tend to get closer to the true (theoretical) probability as the number of trials increases.

The mean of the results obtained from a large number of independent random samples converges to the true value, if it exists...



Probability can be written in the forms of fractions, decimals, or percentages.

For our class, we will write probabilities as decimals, **rounded to three decimal places.**

Example 1) A bag contains 10 marbles – 2 black, 3 blue, 1 red, and 4 white.

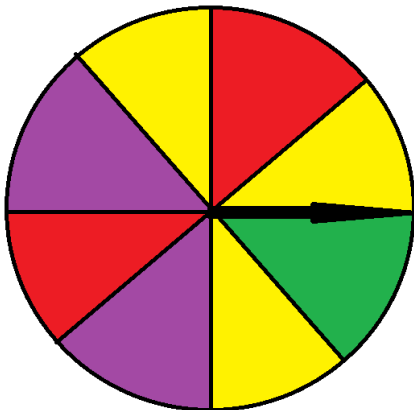
a) What is the probability you will pull out a white marble?

b) What is the probability you will pull out a blue marble?

$$P(\text{white}) = 4/10 = 0.40$$

$$P(\text{blue}) = 3/10 = 0.30$$

Example 2) In the spinner below, each wedge is equal in area.



a) What is the probability that the spinner will land on a red wedge?

$$P(\text{red}) = 1/4 = 0.25$$

b) What is the probability that the spinner will land on a yellow wedge?

$$P(\text{yellow}) = 3/8 = 0.375$$

Probability of a Complement

The probability that an event DOES NOT occur is 1 – probability it does occur or in symbols:

$$P(\text{NOT } A) = 1 - P(A)$$

The notation for “Not A” is A^C , therefore we say: $P(A^C) = P(A') = 1 - P(A)$

Sometimes, it is easier to compute the complement of an event, than the event itself.

Example 3) A coin is flipped three times and the results of each flip noted.

a) Draw the sample space for this event.

$$\Omega = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$$

b) What is the probability a series of three flips will produce exactly one head?

$$P(\text{one H}) = 3/8 = 0.375$$

c) What is the probability that a series of three flips will produce at least one head?

$$P(\text{at least one H}) = 7/8 = 0.875$$

(Frequentist Perspective)

Remember, probability answers the question, what would happen if I did this many, many times?

The myth of the “law of averages”:

- The idea of probability is that randomness is predictable in the long run
- Probability does not allow us to make short-run predictions (Gambler's Fallacy)
- Probability tells us random behavior evens out in the long run. **Future outcomes are not affected by past behavior.**
- Probability does not tell use that it evens out in the short run
 - “You’ve flipped a coin 10 times and it was heads every time! It will be more likely that you get a tail next flip.” → FALSE!!

Probability and Frequency Tables

We saw frequency tables back in Unit 2, when we discussed two categorical variables. We can use frequency tables to find and interpret probabilities.

Example 4) A group of 50 high school students were interviewed and asked what their favorite leisure activity is. The results are in the table below. Use the table to find and interpret the following probabilities.

	Dance	Watch/Play Sports	Read	TOTAL
Male	2	10	8	20
Female	16	6	8	30
TOTAL	18	16	16	50

a) Of all the individuals surveyed, what is the probability of picking a male whose favorite leisure activity is dance?

$$P(\text{male} \cap \text{dance}) = 2/50 = 0.04$$

b) Of all the individuals surveyed, what is the probability of picking a person whose favorite leisure activity is reading?

$$P(\text{read}) = 16/50 = 0.32$$

c) Of all the individuals surveyed, what is the probability of picking someone who does not consider dance their favorite leisure activity?

$$P(\text{dance}') = 1 - 18/50 = 0.64$$

d) Of all the females surveyed, what is the probability of choose a female who likes reading?

$$P(\text{read} | \text{female}) = 8/30 = 0.267$$

Simulations

The imitation of chance behavior, based on a model that accurately reflects the situation, is called a simulation. Simulations are usually done with a table of random digits, calculator random number generator, or computer software.

For Example: What is the probability that a student gets 6 out of 6 questions correct on a true/false quiz written in Greek? (Assume the exam taker does not know any Greek). Should the instructor be concerned about cheating? Perform a simulation using the four-step process to answer this question.

Required Elements (4 step process):

STATE Identify what question you will be answering Is it possible to guess on a T/F quiz and get 6 out of 6 questions correct?	PLAN Describe the tool you will be using to conduct the simulation. Be clear about what values you are assigning to the outcomes, what you are doing each time you are performing a trial, and what you are recording. When guessing on a T/F quiz, we would expect the student to get 3 out of 6 questions correct. Using a random number generator (TI-84), I will assign the values "1" for getting the question correct and "0" for getting the question incorrect. I will randomly generate 6 numbers (0 or 1) and count how many are labeled "correct". This will count as one trial and I will do this for 20 trials.														
DO Record the simulation data (summarize the data if you have a large number of trials) <table><tr><th>n correct</th><th>n trials</th></tr><tr><td>1</td><td></td></tr><tr><td>2</td><td></td></tr><tr><td>3</td><td></td></tr><tr><td>4</td><td></td></tr><tr><td>5</td><td></td></tr><tr><td>6</td><td></td></tr></table>	n correct	n trials	1		2		3		4		5		6		CONCLUDE Answer the question that you asked in the "State" part of the problem, and comment on if these were the results you expected or if they are surprising in some way.
n correct	n trials														
1															
2															
3															
4															
5															
6															